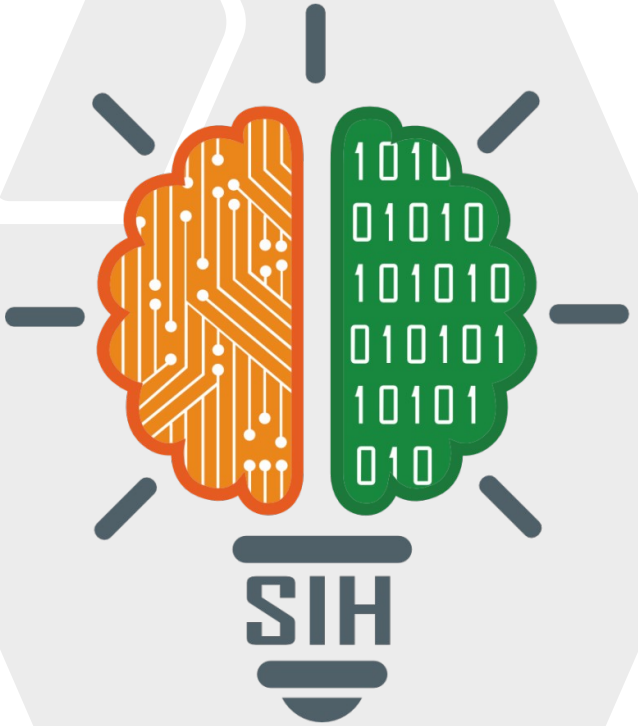
SMART INDIA HACKATHON 202

SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID – 26136**
- **Problem Statement Title– Startup-Friendly Public Procurement Mechanism for Govt. Depts.**
- **Theme– Miscellaneous**
- **PS Category- Software**
- **Team ID-**
- **Team Name- The Optimists**



The Optimists

IDEA TITLE

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Proposed Solution (Describe your Idea/Solution/Prototype) • Detailed explanation of the proposed solution • How it addresses the problem • Innovation and uniqueness of the solution

StartupSetu

Discover innovation. Prove impact. Scale what works.

THE REAL PROBLEM

Departments know the outcome they want, but not **which startup, how to pilot safely, or how to prove it worked** before buying.

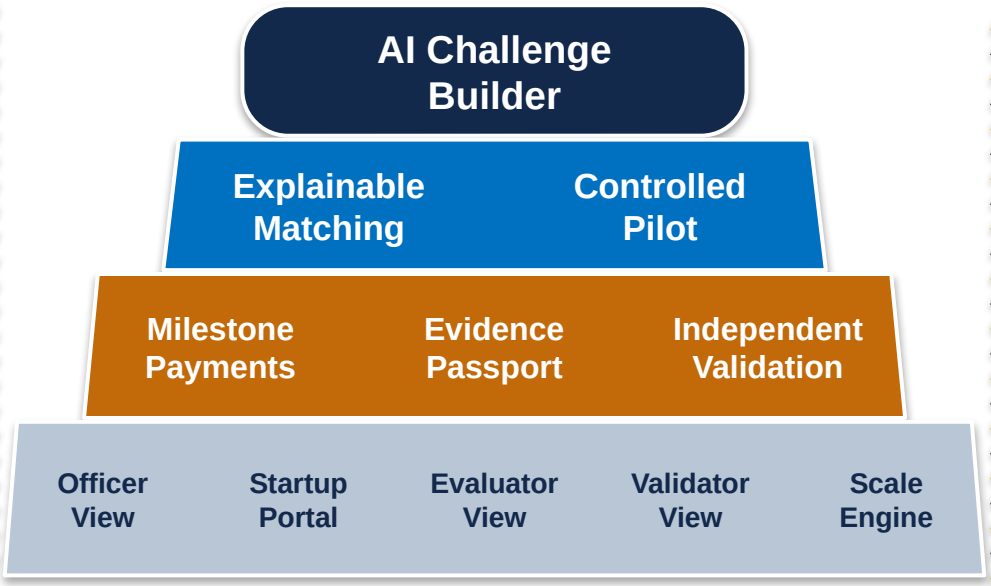
WHY IT MATTERS

Conventional procurement suits standardised vendors — **not early-stage innovation**. Good startups give up on long, opaque sales cycles.

OUR SOLUTION

An **innovation lifecycle layer** around GeM/procurement: structured challenge → explainable match → controlled pilot → independent proof → procurement.

WHAT STARTUPSETU IS BUILT FROM



AI recommends at every layer — the officer always makes the final call.

RISK → SOLUTION

Guesswork on who can solve it	Explainable Startup Matching
Risky statewide rollout, unproven tech	Controlled, KPI-Driven Pilot
No proof before big purchase	Independent Validation + Evidence Passport

Not another GeM marketplace — StartupSetu is the innovation lifecycle layer around procurement.

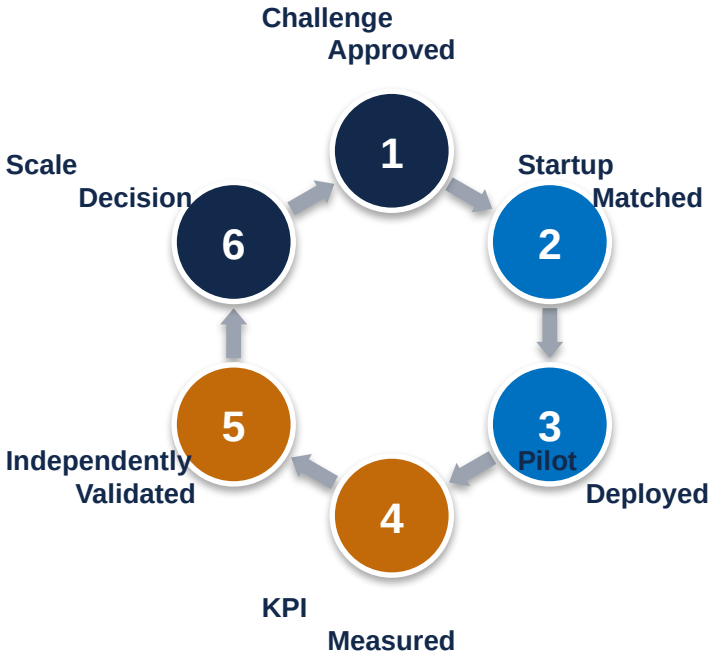
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TECHNICAL APPROACH

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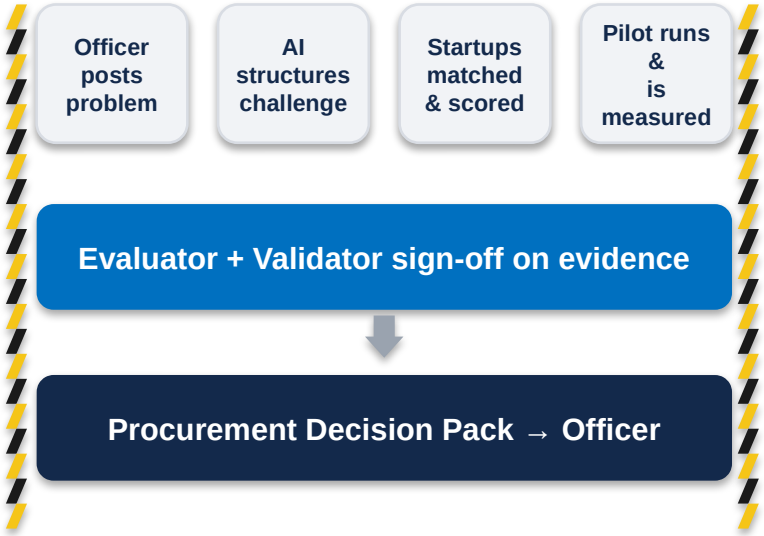
Technologies to be used (e.g. programming languages, frameworks, hardware) • Methodology and process for implementation (Flow Charts/Images/ working prototype)

PILOT LIFECYCLE (METHODOLOGY)



Each turn of the cycle is one funded, time-boxed pilot.

HOW A CHALLENGE FLOWS



Milestone-gated payments run alongside this same flow: Deploy → Validate → Achieve KPI → Final Review.

TECHNOLOGIES USED

Frontend
React + Tailwind CSS

Backend
FastAPI / Node.js microservices

Data
PostgreSQL + pgvector (semantic search)

AI Layer
LLM for structuring, matching, evidence review

Security
RBAC, encryption, tamper-evident audit trail

Integration
Adapters for GeM & govt/startup APIs

Hybrid matching = eligibility rules + embeddings + explainable scoring.

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FEASIBILITY AND VIABILITY

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✓

Analysis of the feasibility of the idea

– **Software-first** — no new hardware; deploys on standard govt. cloud/on-prem infra.

– **Modular APIs** let departments integrate at their own pace.

– **Pilot-first adoption** — small, reversible pilots before commitment.

– **Human-controlled decisions** at every approval gate.

– **Real precedent** — OECD-documented PPI models and GeM's own Startup Runway already work this way.

⚠

Potential challenges and risks

– **Policy variation** across departments/states in procurement rules.

– **Biased AI ranking** if training data skews toward larger, established startups.

– **Startup data quality** and integration gaps with legacy systems.

– **Change management** — departments must trust a new discovery channel over habit-driven vendor lists.



Strategies for overcoming these challenges

– **Configurable workflows** adapt to each department's own policy.

– **Explainable scoring** + regular fairness audits of the matching model.

– **Human validation** and adapter-based integrations for legacy systems.

– **Start narrow** — pilot in 1–2 departments first, mirroring Amsterdam's Startup-in-Residence model.

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IMPACT AND BENEFITS

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Potential impact on the target audience • Benefits of the solution (social, economic, environmental, etc.)

Reduced Guesswork
Explainable matching replaces manual vendor discovery.

Lower Pilot Risk
Small, KPI-bound pilots before any big spend.

Faster Time-to-Value
Structured challenges shorten sourcing cycles.

Audit-Ready Trail
Every decision is backed by stored evidence.

IMPACT | BENEFIT

Government
Lower-risk, audit-ready procurement decisions.

Startups
Real demand visibility + milestone payments.

Citizens
Faster access to solutions that actually work.

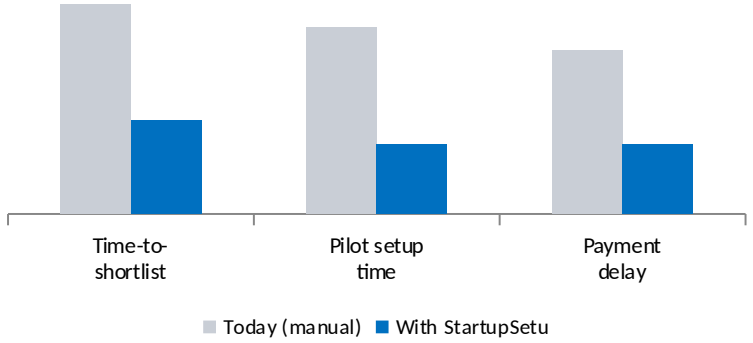
Usability
Guided, role-based, multilingual-ready platform.

WHAT CHANGES

Today: ad-hoc pilots, no comparable proof

With StartupSetu: evidence-backed procurement

ILLUSTRATIVE COMPARISON — CONCEPTUAL TARGETS, NOT MEASURED DATA



Pilot-to-Scale Conversion
directional aim ↑

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RESEARCH AND REFERENCES

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Details / Links of the reference and research work

Existing: GeM / Startup India
Helps transact — once you know the vendor.

StartupSetu: Challenge → Match → Pilot → Evidence → Scale
Helps discover & prove — before that decision exists.

KEY REFERENCES

Frames procurement as a genuine innovation-policy

OECD PPI Report, 2024
LINK

PPI Innovation Policy Review
LINK

What design choices make procurement spur innovation

Amsterdam's real-world challenge-based startup

Startup-in-Residence Study
LINK

GeM Startup Runway
LINK

India's own startup procurement runway, live on GeM

Why turnover & prior-experience rules block

Startup India Procurement Policy
LINK

Edquist et al., PPI Foundations
LINK

The foundational academic case for innovation procurement

ECOSYSTEM MAP

Startup India / Startup Databases → **StartupSetu** → **Pilot & Independent Validation** → **Government Procurement Ecosystem (GeM & Depts.)**

CONCEPT UI SKETCH (WIREFRAME — NOT A LIVE PRODUCT)

